Wonseon Lim

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Education

2021 - present **Ph.D. Candidate**, Computer Science and Engineering at **Chung-Ang University**
Advisor: Prof. Jaesung Lee, Prof. Dae-Won Kim

2024 - 2024 **Visiting Student**, Mechanical and Industrial Engineering at **University of Toronto**
Advisor: Prof. Chi-Guhn Lee

2019 - 2021 **Master's Degree**, Computer Science and Engineering at **Chung-Ang University**
Advisor: Prof. Jaesung Lee, Prof. Dae-Won Kim

2013 - 2019 **Bachelor's Degree**, Applied Statistics at **Chung-Ang University**

Research Interests

My research focuses on designing **efficient and adaptive deep learning systems for on-device environments**, where models must operate under strict constraints on computation, memory, and latency. I am particularly interested in bridging **learning algorithms** and **deployment-aware optimization** to enable scalable, deployable AI across diverse hardware platforms.

My current research interests include:

- **On-Device AI**: On-Device Training, Online Continual Learning, Token Pruning
- **Model Optimization**: Hardware-Aware Neural Architecture Search, Post-Training Quantization

Publications

[1] H. Im*, **W. Lim***, D. Kim. "MAND: Modality-Aware Novelty Detection for Open-World Egocentric Activity Recognition." ICML 2026 Workshop on Epistemic Intelligence in Machine Learning.

[2] **W. Lim**, J. Lee, D. Kim. "Critical Patch-Aware Sparse Prompting with Decoupled Training for Continual Learning on the Edge." The IEEE/CVF Computer Vision and Pattern Recognition Conference (CVPR) 2026.

[3] **W. Lim**, Y. Zhou, D. Kim, J. Lee. "MixER: Mixup-based Experience Replay for Online Class-Incremental Learning." *IEEE Access*, March 2024

[4] J. Park, **W. Lim**, D. Kim, J. Lee. "GTSNet: Flexible architecture under budget constraint for real-time human activity recognition from wearable sensor." *Engineering Applications of Artificial Intelligence*, Sep 2023

[5] **W. Lim**, W. Seo, D. Kim, J. Lee. "Efficient Human Activity Recognition using Lookup Table-based Neural Architecture Search for Mobile Devices." *IEEE Access*, July 2023

[6] W. Seo, J. Park, **W. Lim**, D. Kim, J. Lee. "Achieving an optimal group structure in a neural architecture search." *Electronics Letters*, July 2022

[7] J. Lee, W. Seo, J. Park, **W. Lim**, J. Oh, N. Moon, J. Lee. "Neural Network-based Method for Diagnosing Graves' Orbitopathy using Orbital Computed Tomography." *Scientific Reports*, July 2022

[8] J. Park, **W. Lim**, D. Kim, J. Lee. "Multi-Temporal Sampling Module for Real-Time Human Activity Recognition." *IEEE Access*, May 2022

Project Experiences

Sustainable Learning Machines for On-Device Holistic Intelligence *2023 – Present*

- **Techniques:** On-Device Training [2], Online Continual Learning [3], Hardware-Aware NAS [4], Multimodal Learning [1], Out-of-Distribution Detection [1]
- Related Link: github.com/laymond1/cps-prompt, github.com/laymond1/MixER, github.com/jgpark92/GTSNet

Development of Target Recognition No-Code Neural Network Creation and Operational Framework (TANGO) *2022 – 2023*

- **Techniques:** YOLO, Hardware-Aware NAS, Post-Training Quantization, Docker
- Related Link: github.com/ML-TANGO/TANGO
- TANGO Community Conference: TANGO/AutoNN

Deep Learning Self-assembly Framework for On-Device AI *2020 – 2022*

- **Techniques:** Hardware-Aware NAS ([5],[6]), Real-time Human Activity Recognition ([5],[8]), Multi-scale Temporal Sampling [8]
- Related Link: github.com/laymond1/HAR-Model-Latency, github.com/minercode625/grunas, github.com/jgpark92/MTS-ConvNet

Emotion-specialized Text-to-Video Retrieval *Jan. 2024 – Apr. 2024*

- **Techniques:** Vision-Language Model, Cross-Modal Learning, Sentiment Analysis
- Related Material: PPT

Optimized Molecular Generation with Molecular Representations *Mar. 2024 – Jun. 2024*

- **Techniques:** Variational Autoencoder (VAE), Bayesian Optimization, Latent Space Analysis
- Related Material: PPT
- Related Link: github.com/sunghwanism/NewMoses

3D Medical Image Analysis for Objective Assessment of Orbital Disease *2019 – 2021*

- **Techniques:** Orbit Segmentation, 3D Image Classification
- Related Paper: [7]
- Related Link: github.com/laymond1/Graves-Orbitopathy-NN

Clinical Deep Learning–based Image Analysis for Gallbladder Polyp and Wall Thickening *2020 – 2021*

- **Techniques:** Polyp Segmentation, Medical Image Analysis

*Equal contribution.

Honors and Awards

2024	**MSIT International Research Training Fellowship** Funded by the Ministry of Science and ICT (MSIT)
2021–2023	**Graduate Research Scholarship** Funded by Chung-Ang University
2021	**Top 10% Finish**, CommonLit Readability Prize Kaggle Competition
2018	**Grand Prize**, Self-Directed Creative Credit Program Statistical analysis of fine dust data, Chung-Ang University
2017	**Encouragement Award**, University Student Data-Driven Tourism Marketing Contest Marketing strategy for the Middle East and Central Asia, Korea Tourism Organization

Academic Service

- IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) *2026*
- Association for the Advancement of Artificial Intelligence (AAAI) *2026*

Skills

Certifications	Advanced Data Analytics Semi-Professional (ADsP), OPIc(IH)
Languages	English: Intermediate, Korean: Native
Techniques	Python, C++, SQL, R, LaTeX, Git, Docker
Machine Learning Tools	PyTorch, Tensorflow, Pytorch-Mobile, Tensorflow-Lite, ONNX, Pandas, scikit-learn